

# Zane Kun Li

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## Affiliation

08/2023 - Present Assistant Professor, North Carolina State University  
05/2024 - 06/2024 Visiting Scholar, School of Mathematics & Statistics, University of Sydney  
08/2022 - 08/2023 RTG Postdoctoral Fellow, University of Wisconsin-Madison  
08/2019 - 07/2022 NSF & Zorn Postdoctoral Fellow, Indiana University Bloomington

## Education

09/2013 - 06/2019 Ph.D, Mathematics, University of California, Los Angeles  
Advisor: Terence Tao  
09/2009 - 06/2013 A.B. magna cum laude, Mathematics, Princeton University

## Grants

2025-2030 Simons Travel Support for Mathematicians SFI-MPS-TSM-00013946, PI (\$37635)  
2025-2028 NSF Award DMS-2453448  
*Extensions and Applications of Fourier Decoupling Theory*, PI (\$226465)  
2022-2025 NSF Award DMS-2409803 (formerly DMS-2154531 and DMS-2311174)  
*Decoupling theory and exponential sum estimates*, PI (\$102063)

## Fellowships

2019-2022 NSF Mathematical Sciences Postdoctoral Research Fellowship, DMS-1902763  
2018-2019 Girsky Fellowship Award, Department of Mathematics, UCLA  
2013-2018 NSF Graduate Research Fellowship

## Journal Articles

1. Anthony Carbery, Zane Kun Li, Yixuan Pang, Po-Lam Yung, *A weighted formulation of refined decoupling, and inequalities of Mizohata–Takeuchi-type for the moment curve*, **Journal of Geometric Analysis**, 36 (2026), article number 48.
2. Christopher Boyer and Zane Kun Li, *An improved example for an autoconvolution inequality*, to appear in *Experimental Mathematics*.
3. Shival Dasu, Hongki Jung, Zane Kun Li, José Madrid, *Mixed norm  $l^2$  decoupling for paraboloids*, **International Mathematics Research Notices**, Vol. 2023, no. 20, 17972-18000.
4. Brian Cook, Kevin Hughes, Zane Kun Li, Akshat Mudgal, Olivier Robert, and Po-Lam Yung, *A decoupling interpretation of an old argument for Vinogradov’s Mean Value Theorem*, **Mathematika** 70 (2024), no.1, e12231.
5. Shaoming Guo, Zane Kun Li, and Po-Lam Yung, *Improved discrete restriction for the parabola*, **Mathematical Research Letters**, 30 (2023), no. 5, 1375-1409.
6. Alan Chang, Jaume de Dios Pont, Rachel Greenfeld, Asgar Jamneshan, Zane Kun Li, and José Madrid, *Decoupling for fractal subsets of the parabola*, **Mathematische Zeitschrift** 301 (2022), 1851-1879.

7. Shaoming Guo, Zane Kun Li, Po-Lam Yung, and Pavel Zorin-Kranich, *A short proof of  $l^2$  decoupling for the moment curve*, **American Journal of Mathematics** 143 (2021), no. 6, 1983–1998.
8. Shaoming Guo, Zane Kun Li, and Po-Lam Yung, *A bilinear proof of decoupling for the cubic moment curve*, **Transactions of the American Mathematical Society** 374 (2021), no. 8, 5405–5432.
9. Zane Kun Li, *An  $l^2$  decoupling interpretation of efficient congruencing: the parabola*, **Revista Matemática Iberoamericana** 37 (2021), no. 5, 1761–1802.
10. Zane Kun Li (with an appendix by Jean Bourgain and Zane Kun Li), *Effective  $l^2$  decoupling for the parabola*, **Mathematika** 66 (2020), no. 3, 681–712.
11. Zane Kun Li, *Decoupling for the parabola and connections to efficient congruencing*, Ph.D. thesis, 2019, available at <https://escholarship.org/uc/item/0cz3756c>.
12. Zane Kun Li, *Quadratic twists of elliptic curves with 3-Selmer rank 1*, **International Journal of Number Theory** 10 (2014), no. 5, 1191–1217.
13. David Corwin, Tony Feng, Zane Kun Li, and Sarah Trebat-Leder, *Elliptic curves with full 2-torsion and maximal adelic Galois representations*, **Mathematics of Computation** 83 (2014), no. 290, 2925–2951.
14. Zane Kun Li and Alexander W. Walker, *Arithmetic properties of Picard-Fuchs equations and holonomic recurrences*, **Journal of Number Theory** 133 (2013), no. 8, 2770–2793.
15. Zane Kun Li, *A normal form for cubic surfaces*, **International Journal of Algebra** 4 (2010), no. 5, 233–239.
16. Zane Kun Li, *On a special case of the intersection of quadric and cubic surfaces*, **Journal of Pure and Applied Algebra** 214 (2010), no. 11, 2078–2086.
17. Stephen P. Humphries and Zane Kun Li, *Counting powers of words in monoids*, **European Journal of Combinatorics** 30 (2009), no. 5, 1297–1308.

## Conference Proceedings

1. Jianhui Li, Zane Kun Li, Po-Lam Yung, *Strichartz inequalities: some recent developments*, arXiv:2310.15306, to appear in *From Classical Analysis to Analysis on Fractals, A Tribute to Robert Strichartz*, Volume 2.
2. Zane Kun Li, *An introduction to decoupling and harmonic analysis over  $\mathbb{Q}_p$* , **Contemporary Mathematics** 792 (2024), 67–94.

## Invited Seminar Talks

|      |         |                                               |                                                                              |
|------|---------|-----------------------------------------------|------------------------------------------------------------------------------|
| 2026 | June    | National Taiwan Normal University             | Nonlinear Analysis Seminar                                                   |
|      | June    | Southern University of Science and Technology |                                                                              |
|      | June    | Monash University                             | Analysis, PDE and Geometry Seminar                                           |
|      | May     | Australian National University                | PDE and Analysis Seminar                                                     |
|      | April   | Virginia Tech                                 | Analysis and Mathematical Physics Seminar                                    |
|      | April   | University of Kansas                          | (virtual) Diff. Equations, Dynamical Systems, and Geometric Analysis Seminar |
|      | January | Seoul National University                     | Analysis Seminar                                                             |
| 2025 | October | University of South Carolina                  | Quantum Information/Analysis Seminar                                         |
|      | May     | University of Rochester                       | Combinatorics Seminar                                                        |

|      |           |                                      |                                              |
|------|-----------|--------------------------------------|----------------------------------------------|
|      | April     | University of Mississippi            | Number Theory Seminar                        |
|      | April     | National Taiwan Normal University    | (virtual) Nonlinear Analysis Seminar         |
|      | February  | Auburn University                    | Seminar on Analysis and Stochastic Analysis  |
| 2024 | December  | National Taiwan University           | NCTS Nonlinear PDE and Analysis Seminar      |
|      | June      | Monash University                    | Analysis Seminar                             |
|      | June      | Australian National University       | PDE & Analysis Seminar                       |
|      | May       | University of New South Wales        | Pure Mathematics Seminar                     |
|      | May       | UCLA                                 | Analysis Seminar                             |
|      | April     | Virginia Tech                        | Analysis and Mathematical Physics Seminar    |
|      | March     | UNC Chapel Hill                      | Analysis Seminar                             |
|      | January   | North Carolina State University      | Differential Eqns/Nonlinear Analysis Seminar |
| 2023 | October   | Washington University of St. Louis   | Analysis Seminar                             |
|      | October   | Duke University                      | Applied Math & Analysis Seminar              |
|      | July      | Australian National University       | Colloquium                                   |
|      | July      | University of New South Wales        | Number Theory Seminar                        |
|      | February  | University of California, Santa Cruz | Colloquium                                   |
| 2022 | November  | University of Georgia                | Colloquium                                   |
|      | November  | North Carolina State University      | Colloquium                                   |
|      | September | University of Wisconsin-Madison      | Analysis Seminar                             |
|      | May       | University of California, Riverside  | Colloquium                                   |
|      | March     | –                                    | Virtual Harmonic Analysis Seminar            |
|      | March     | Australian National University       | (virtual) Analysis Seminar                   |
|      | February  | Caltech                              | Discrete Analysis Seminar                    |
|      | February  | UCLA                                 | Analysis Seminar                             |
|      | January   | Caltech                              | (virtual) Discrete Analysis Seminar          |
| 2021 | November  | Shandong University                  | (virtual) Number Theory Seminar              |
|      | October   | University of Kansas                 | (virtual) Analysis Seminar                   |
|      | April     | MIT                                  | (virtual) PDE/Analysis Seminar               |
|      | March     | Indiana University                   | (virtual) Analysis Seminar                   |
| 2020 | December  | –                                    | (virtual) Chinese Webinar on APDE            |
|      | February  | Caltech                              | UCLA/Caltech Joint Analysis Seminar          |
|      | February  | University of Chicago                | Calderón-Zygmund Analysis Seminar            |
|      | February  | UW Madison                           | Analysis Seminar                             |
|      | January   | Indiana University                   | Analysis Seminar                             |
| 2019 | November  | UIUC                                 | Harmonic Analysis & Diff Eq Seminar          |
|      | October   | University of Rochester              | Combinatorics Seminar                        |
|      | October   | Purdue University                    | Analytic NT & Harmonic Analysis Sem.         |
|      | February  | UC Davis                             | PDE and Applied Math Seminar                 |
|      | January   | University of British Columbia       | Harmonic Analysis Seminar                    |
| 2018 | December  | Chinese University of Hong Kong      | Two one hour talks                           |
|      | December  | University of Bristol                | Analysis & Geometry Seminar                  |
|      | May       | Caltech                              | Analysis Seminar                             |
|      | May       | UCLA                                 | Analysis Participating Seminar               |

### Invited Conference Talks

|      |          |                                                                                  |
|------|----------|----------------------------------------------------------------------------------|
| 2026 | October  | AMS Fall Eastern Sectional – Special Session on Harmonic Analysis & PDEs         |
|      | July     | Philadelphia Harmonic Analysis and Differential Equations (PHADE 2026)           |
|      | February | Harmonic Analysis and its Interactions @ Nagoya 2026                             |
| 2025 | March    | 16th Nagoya Workshop on Differential Equations                                   |
|      | February | 4th Harmonic Analysis Workshop in Seoul                                          |
| 2024 | May      | MATRIX Harmonic Analytic Connections                                             |
|      | April    | AMS Spring Eastern Sectional – Special Session on HA & their Applications to PDE |
|      | March    | Pittsburgh Links among Analysis and Number Theory (PLANT)                        |

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|------|-----------|------------------------------------------------------------------------------------|
| 2023 | September | NC State College of Sciences Excellence Symposium                                  |
|      | April     | AMS Spring Central Sectional – Special Session on GMT and HA                       |
|      | March     | Harmonic Analysis and Fractal Sets 2023                                            |
|      | March     | AMS Spring Southeastern Sectional – Special Session on Harmonic Analysis           |
| 2022 | June      | Fourier Analysis @200 – Young Researchers Symposium                                |
|      | March     | (virtual) AMS Spring Central Sectional – Special Session on Harmonic Analysis      |
| 2021 | November  | (virtual) The 17th Prairie Analysis Seminar                                        |
|      | August    | (prerecorded short talk) HIM Trimester Program Harmonic Analysis & Analytic NT     |
|      | March     | (prerecorded) Fourier restriction online 2021                                      |
|      | February  | (virtual) AIM workshop: Arithmetic Stat., Discrete Restriction, & Fourier Analysis |
| 2020 | December  | (virtual) Canadian Math. Society Winter Meeting – Session on Discrete Analysis     |
|      | August    | (virtual) The Eighth Pacific Rim Conference in Mathematics                         |
| 2019 | September | AMS Fall Central Sectional – Special Session on Recent Dev. in Harmonic Analysis   |
|      | June      | (Heilbronn Inst., Bristol) Efficient Cong. & Decoupling Focused Research Workshop  |
|      | January   | (JMM) AMS Special Session on Counting Methods in Number Theory                     |
|      | January   | (JMM) AMS Special Session on HA: Recent Dev. on Oscillatory Integrals              |
| 2017 | October   | HCM Kopp Summer School on Decoupling & Polynomial Methods in Analysis              |

## Invited Lecture Series

|      |         |                                 |                                              |
|------|---------|---------------------------------|----------------------------------------------|
| 2026 | August  | Nanjing University              | 9.5 hour lecture series on decoupling        |
| 2025 | March   | University of Göttingen         | RTG2491 Spring School, 4 hour lecture series |
| 2023 | July    | Australian National University  | 4.5 hour lecture series on decoupling        |
| 2022 | Apr-May | NCTS/National Taiwan University | 10 hour lecture series on decoupling         |
| 2020 | Oct-Nov | Discrete Analysis Working Group | (virtual) Four one hour talks on decoupling  |

## Service

- Co-organizer of NC State’s Stochastics/Discrete Analysis Seminar, Fall 2024 to present
- Co-organizer of AIM Restriction Community on Fourier Restriction and Related Problems, Fall 2024 to present
- Co-organizer of OARS (Online Analysis Research Seminar), Fall 2020 to present
- Co-organizer of the following conferences:
  - NCTS Workshop on Harmonic Analysis, Taipei, TW, December 2024
  - 2025 AMS Spring Central Sectional Meeting, Special Session on Recent Trends in Harmonic Analysis and PDE, Lawrence, KS, March 2025
  - Summer School on “Discrete directions in harmonic analysis”, Kopp, DE, October 2025
- Referee for Analysis & PDE, Essential Number Theory, Geometric and Functional Analysis, IMRN, Journal d’Analyse Mathématique, Journal of Fourier Analysis and Applications, Journal of Functional Analysis, Mathematische Zeitschrift, Mathematische Annalen, Pacific Journal of Mathematics, Proceedings of the Edinburgh Mathematical Society, Revista Matemática Iberoamericana
- Reviewer for Math Reviews

(last updated August 7, 2026)